

Problem Statement

Project: Job Market Reality Check

The demand for Data Analysts is growing rapidly, but job seekers often struggle to identify which skills, technologies, locations, and industries provide the best career opportunities. Job postings contain valuable information, but the data is unstructured and difficult to analyze manually.

This project aims to build an end-to-end data analytics solution that cleans and transforms real-world job posting data, extracts technical skills from job descriptions, stores the processed data in PostgreSQL, and analyzes hiring trends, salary patterns, and in-demand skills using SQL, Excel, and Power BI.

Objectives

- Clean and preprocess job posting data using Python and Pandas.
- Extract key skills from job descriptions using Regular Expressions.
- Store cleaned data in PostgreSQL.
- Analyze salary trends, skill demand, and geographic hiring patterns using SQL.
- Create interactive dashboards in Power BI for business insights.
- Help students and job seekers understand current market requirements and make data-driven career decisions.

Expected Outcome

A complete analytics pipeline that converts raw job postings into meaningful insights, highlighting the most in-demand skills, salary drivers, and hiring trends to support informed career planning.